On the number line we first start at zero facing positive direction and move 7 steps backward, reaching –7. Then we move 9 steps backward to represent –9 and reach at –16. Thus, we get (–7) + (–9) = –16.

Example 1.2

Add (i) (–40) and (30) (ii) 60 and (–50)

Solution (i) (–40) and (30)

–40 + 30 = –10

(ii) 60 and (–50)

60 + (–50) = 60 – 50 = 10

Example 1.3

Add: (i) (–70) and (–12) (ii) 103 and 39.

Solution (i) (–70) + (–12) = –70 – 12 = – 82

(ii) 103 + 39 = 142

Example 1.4

A submarine is at 32 feet below the sea level. Then it moves up 8 feet. Find the depth of the submarine.

Solution A submarine is 32 feet below sea level.

Therefore, it is represented by –32

Next it moves up 8 feet.

Moves above is represented as +8

The depth of the submarine = –32 + 8 = –24

Therefore, the submarine is located at 24 feet below the sea level.

Example 1.5

Sita saved ₹ 225.00 and she has spent ₹ 400 on credit basis for the purchase of stationery. Find her due amount.

Solution The amount Sita has ₹ 225

The amount spent for stationery on credit = ₹ 400

The due amount to be paid = 225 – 400 = – 175

Therefore, Sita has to pay ₹ 175

Example 1.6

From the ground floor a man went up six floors and came down six floors. In which floor is he now?

Solution Starting point = Ground floor

Number of floors climbed up = +6

Number of floors climbed down = –6

Now the landing point = +6 – 6 = 0 (ground floor)

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